

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA, 14.15 kB circular plasmid which contains a part of pSV2neo.
Species Used	Mouse, SP2/0, myeloma [Sp-2]		

Before the Pulse

Cell Growth Medium	10% Fetal Bovine Serum /RPMI 1640 (GIBCO/BRL,Sigma)	Growth Phase at Harvest	5.4 X 10 (5) cells / ml
		Pre-pulse Incubation	on ice for 10 minutes

Wash Solution	Phosphate Buffered Saline, 2 times
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The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	21 °C		
Electroporation Medium*	Phosphate Buffered Saline	Cuvette Gap	0.4 cm
Cell Density	1.25 x 10 (7) cell / ml	Voltage	0.22 kV
Volume of Cells	800 µl	Field Strength	0.55 kV/cm
DNA Concentration	1.12 µg / µl		
DNA Resuspension Buffer	0.1 x TE buffer (1x=10 mM Tris, 1mM EDTA, pH 8.0)	Capacitor	960 µF
Volume of DNA	18 µl	Resistor	(Pulse Controller) Ω none
		Time Constant	18.5 msec

After the Pulse	
Outgrowth Medium	10% Fetal Bovine Serum / RPMI 1640

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

- *FEBS Lett.* **244**:301 (1989). Y. Kurosawa *et. al.* Convenient plasmid vectors for construction of chimeric mouse/human antibodies.
- *Cancer Research* **50**: 3167 (1990). T. Tsuruo *et. al.* Mouse-Human Chimeric Antibody against the multidrug transporter P-Glycoprotein

PBS: 1x = 8g NaCl,0.2g KCl,0.2g KH₂PO₄, 1.15g Na₂HPO₄

Outgrowth Temperature	37 °C
Length of Incubation	48 hours
Selection Method or Assay Used	800 µg / ml of G418
Electroporation Efficiency	50 µg DNA
Per Cent Survival	50%

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Survey Number

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